Postoperative Adrenal Hemorrhage

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Twelve elderly patients without Waterhouse-Friderichsen syndrome or adrenal tumor who had spontaneous adrenal hemorrhage one to 33 days after operation are described. Seven of these patients had operations on the gastrointestinal tract, one on the biliary system, two on the genitourinary system and two on the central nervous system. Important factors relating to adrenal hemorrhage included: intra-abdominal sepsis in 5 patients, cancer in 4, pneumonia in 4, coagulation defects in 2, exogenous steroids in 2, and syphilis in one patient. Spontaneous adrenal hemorrhage should be considered in patients whose condition deteriorates rapidly after operation and in whom no other explanation is plausible. Its detection and appropriate therapy can be lifesaving.

SPONTANEOUS ADRENAL HEMORRHAGE is an unusual postoperative complication which may result in the patient's unexpected rapid deterioration or death. The clinical manifestations of this condition are often subtle and are usually attributed to other postoperative problems. The diagnosis, therefore, is infrequently made. This is most unfortunate because, if diagnosed, adrenal hemorrhage may be successfully treated. The purpose of this retrospective study is to draw attention to this frequently fatal condition and to review some of the predisposing conditions with which adrenal hemorrhage is associated.

Clinical Manifestations

At the University of California Hospitals, San Francisco, from 1960 to 1973, evidence of postoperative adrenal hemorrhage unassociated with Waterhouse Friderichsen syndrome, documented bacteremia, or adrenal tumor was found in 12 patients at post-mortem examination. In 5 of these patients adrenal hemorrhage was the principal cause of death and in other patients it was an important contributing factor.

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Two brief case reports are described to emphasize the fact that in some postoperative patients, as in Case I, the diagnosis of adrenal hemorrhage should be made or at least considered if one is aware of the various clinical manifestations of adrenal hemorrhage. As in Case II, no pre-mortem symptoms or signs of adrenal hemorrhage are apparent.

Case Reports

Case 1. A 61-year old Caucasian man was hospitalized with a benign gastric ulcer which unfortunately, perforated at time of gastroscopy. Pertinent contributing history was the fact that the patient had been taking 7.5 mg of Prednisone daily for asthma. At operation a benign perforated gastric ulcer was closed without difficulty and the postoperative course was complicated by an interstitial pneumonia. The pneumonia appeared to be resolving when on the seventeenth postoperative day the patient suddenly appeared gravely ill and developed a fever of 39.3 C, mild confusion, and hypotension. An electrocardiogram and chest roentgenogram revealed no acute changes. Five blood cultures showed no growth and the patient failed to respond to intravenous fluid replacement and Aramine. He died on the eighteenth postoperative day and post-mortem examination revealed bilateral adrenal hemorrhage as the probable cause of death. The patient's moribund appearance, unexplained hypotension, fever and confusion were all clinical manifestations of adrenal apoplexy. Spontaneous adrenal hemorrhage should have been considered since no other more common cause of these symptoms and signs such as a pulmonary embolus, septicemia or myocardial infarction was present.

Case 2. An 84-year old Caucasian woman was seen by the urology service for fulguration of a bladder papilloma with past history of surgery 7 years previously for carcinoma of the tongue. At operation the bladder was perforated. This perforation was immediately recognized and repaired and the patient recovered without incident. Two weeks later, there was a second transurethral resection of the bladder tumor. The evening of operation, the patient fell out of bed. This may

have been due to confusion and four hours later she was found dead in bed.

At post-mortem examination bilateral adrenal hemorrhage and adrenal vein thrombosis were found. The adrenal medulla of both glands were replaced by hemorrhagic clot and the cortex was compressed by the expanding hematoma. There was also extension of the hemorrhage into the periadrenal tissues. The pituitary was normal and there was no evidence of residual carcinoma of the tongue.

A summary of the pertinent facts in 12 patients with adrenal hemorrhage are listed in Table 1. Eight of the

TABLE 1. *Postoperative Adrenal Hemorrhage.*

Case	Age,Sex	Operation	Associated Conditions	Time From Operation to Death	Major Clinical and Laboratory Findings	Adrenal Pathology
1,HS	61,M	Gastroscopy with post gastroscopy gastric perforation. Treated by closure of benign gastric perforation	Benign gastric ulcer; resolving interstitial pneumonia; Asthma, (Rx-Prednisone 7.5 mg daily)	18 dys	Acute onset of hypotension confusion and fever (39.3 C) (Negative blood cultures)	Bilateral adrenal hemorrhage
2,PF	84,M	Two transurethral resections for bladder papilloma	Bladder papilloma, status post transurethral fulguration with perforation 15 days prior to death; atherosclerosis; History of carcinoma of tongue	15 dys and 1 dy	Pre-mortem confusion	Acute bilateral hemorrhage with bilateral vein thrombosis and hemorrhage into periadrenal tissue. Pituitary normal
3,BE	81,M	Right colectomy	Carcinoma cecum with extension to bladder; carcinoma prostate; bronchopneumonia	10 dys	Confusion and disorientation pre and postop. Electrolytes normal	Adrenal infarction with adrenal vein thrombosis
4,WK	86,F	Cholecystojejunostomy	Adenocarcinoma of common bile duct with total obstruction	13 dys	Weakness, jaundice and anorexia pre and postop. Electrolytes normal. Febrile prior to demise	Bilateral adrenal necrosis and hemorrhage. Bilateral adrenal vein thrombosis
5,TG	69,M	Gastropexy for hiatal hernia and gastrostomy	Atherosclerosis: S/P resection Zenker diverticulum; P/O wound infection and leakage around gastrostomy	51 dys after Zenker's diverticulectomy, 22 dys after hiatal hernia repair	Sudden development of cyanosis, dyspnea and hypotension	Bilateral adrenal infarction with adrenal vein thrombosis
6,AM	74,M	Transurethral resection prostate	Benign prostatic hypertrophy with postoperative hemorrhage requiring 22 units of blood. Cardiac arrest	16 dys	Pre-mortem shortness of breath and hypotension. Elevated blood urea nitrogen (136 mg/100 ml) creatinine (2.3 mg/100ml)	Bilateral hemorrhage in medulla with extension into cortex. Arterial and venous thrombosis and inflammatory cells in periadrenal fat
7,VP	82,M	Surgical drainage of perforated appendix	Perforated appendix with abscess formation and minimal peritonitis; syphilis with dilated base aorta, and Argyll Robertson pupils; hemorrhagic spleen	2 dys	Anorexia, abdominal pain fever (38.6 C) leukocytosis	Massive hemorrhage with extravasation into periadrenal tissues
8,HE	70,F	Double barrel colostomy	Diverticulitis with perforation and peritonitis; atherosclerosis, pneumothorax	1.5 dys	Anorexia, emesis, abdominal pain, temperature (36.1 C)	Right adrenal hemorrhage. Pituitary normal

Continued

TABLE 1. *Postoperative Adrenal Hemorrhage (continued)*

Case	Age,Sex	Operation	Associated Conditions	Time From Operation to Death	Major Clinical and Laboratory Findings	Adrenal Pathology
9,DP	82,M	Sigmoid resection	Volvulus of sigmoid colon with obstruction; atherosclerosis	11 dys	Premortem hypotension and abdominal pain. Serum potassium ranged from 2.0-5.3mg% "difficult to regulate"	Bilateral adrenal hemorrhage limited to midportion of cortex
10,RE	82,M	Cecostomy	Obstructing villous adenoma of rectosigmoid; chromaphobe adenoma of pituitary gland, congestive heart failure, Hashimoto's thyroiditis	33 dys	Weight loss, leukocytosis and subnormal temperature	Bilateral adrenal hemorrhage with unilateral adrenal vein thrombosis
11,WR	63,F	Laminectomy	Rheumatoid arthritis. Pneumonia (aspiration). Steroids for arthritis. Thyroid adenoma	6 dys	Abdominal distension	Bilateral adrenal hemorrhage left greater than right
12,OG	62,M	Fronto-temporal craniotomy	Small adrenocortical adenoma; carcinoma lung with metastasis to brain; infected craniotomy wound	15 dys	Febrile (40.0 C) elevated blood urea nitrogen (31 mg%)	Bilateral adrenal hemorrhage

patients were men and 4 were women. The mean age was 72 years; the youngest patient was 61 years old and the oldest was 86 years old. Eight of the 12 patients had bilateral adrenal hemorrhage. Seven patients had had operations on the gastrointestinal system and three of these patients had had preoperative intestinal perforation with peritonitis. One patient had a perforated benign gastric ulcer, one had a perforated appendix with abscess formation, and another had a perforated colonic diverticulum. The other gastrointestinal operations were an anterior gastropexy and gastrostomy for a hiatal hernia, a right colectomy for carcinoma of the cecum, a cecostomy for an obstructing villous adenoma of the rectosigmoid and a sigmoid resection for a sigmoid volvulus with obstruction. One patient had a cholecystojejunostomy for biliary obstruction due to adenocarcinoma of the common bile duct. Two patients had complications after urological operations, one following a prostatectomy and the other after fulguration of a bladder papilloma. The first of these patients had a massive hemorrhage requiring 22 units of blood and resulting in cardiac arrest; and the second patient had perforation of the bladder. Another patient had bilateral adrenal hemorrhage after a craniotomy for metastatic carcinoma of the lung with postoperative wound infection. The last patient, who had bilateral adrenal hemorrhage 6 days after a laminectomy, had been taking steroids for arthritis and also had pneumonia.

This group of elderly patients with postoperative adrenal hemorrhage had many other problems that have been described in association with spontaneous adrenal hemorrhage. Intra-abdominal sepsis (5 patients) and intracranial sepsis (1 patient) without positive blood cultures, occurred in 6 cases, bronchopneumonia and tracheobronchitis in 4 patients, carcinoma in 4 patients, coagulation defects in 2 patients, syphilis in 1 patient and 2 patients were taking exogenous adrenal corticosteroids. Associated endocrine problems were present in three patients. One patient had Hashimoto's thyroiditis and a chromophobe adenoma of the pituitary; one patient had a thyroid adenoma; and another had a small unilateral adrenocortical adenoma. The latter patient had bilateral adrenal hemorrhage at post-mortem examination. The association of Hashimoto's thyroiditis and adrenal hemorrhage suggests the possibility of Schimidt's syndrome, an autoimmune disease characterized by hypothyroidism and hypoadrenalism.

The signs and symptoms of postoperative adrenal hemorrhage were difficult to distinguish from the clinical manifestations resulting from operation or preoperative problems. Pain, anorexia, emesis, confusion, hypotension, fever, abdominal distention and cyanosis occurred in many of these patients but these symptoms were masked by their other clinical problems. Similarly, elevation of white cell count and blood urea nitrogen were found in several patients but other causes for these nonspecific increases were present. Unfortunately, since adrenal hemorrhage and acute adrenal insufficiency were not suspected, quantitative eosinophil counts, serial electrolyte and plasma cortisol levels were not determined.

Discussion

Spontaneous adrenal hemorrhage may develop without predisposing conditions but more often occurs in patients subjected to severe stress.[2,16] Extensive adrenal hemorrhage has been described in association with many of the conditions that occurred in the patients in this series, such as gastrointestinal perforation with intra-abdominal sepsis,[11,12] septicemia,[10,26] coagulation disorders,[8,15,18] abdominal or intracranial tumors or injuries,[11,16,32] cancer,[5] administration of steroids,[37] and syphilis,[3] even without the added stress of a surgical procedure. Other important conditions predisposing to adrenal hemorrhage not found in our group of patients are the involution of the adrenal gland in the newborn[19] or in women immediately post partum.[7,22] Of interest is the fact that in our patients postoperative adrenal hemorrhage never occurred after minor surgical procedures, or for that matter without associated illnesses. Spontaneous adrenal hemorrhage, however, has also been reported without known predisposing conditions.[26]

Surprisingly, only 4 other reports of postoperative adrenal hemorrhage have been published.[5,11,12,35] Taylor was the first to draw attention to the association of surgery and adrenal hemorrhage when he reported the case of a 54-year-old woman who died of bilateral adrenal hemorrhage 24 days after a posterior gastroenterostomy and three days after a duodenjejunostomy for duodenal obstruction. This patient also had intestinal diverticulosis, pneumonia and pernicious anemia. She developed the classical symptoms of adrenal hemorrhage after her second operation; emesis, apathy, back pain, fever and eventually coma. "The general picture suggested peritonitis, but the examination of the abdomen did not confirm the diagnosis."[35]

Six of the 23 patients described by Botteri had adrenal hemorrhage after an operation. Two of these patients died after gastric resections, performed for gastric carcinoma and for duodenal ulcer. The latter patient had bronchopneumonia and was taking ACTH and cortisone. Two patients died after operations on the biliary system, one patient died after a splenectomy, and the last patient died after a closed reduction of a dislocated shoulder performed under general anesthesia.[5]

Fox reported unilateral adrenal hemorrhage in 8 patients with gastric cancer who died after major operations on the stomach, and in a ninth patient who died after partial nephrectomy. He attributed the hemorrhage to direct damage to the adrenal gland or to its blood supply during operation because in each patient the adrenal hemorrhage occurred on the same side as the operative procedure. However, 4 of his patients had anastomotic leaks and 3 had bronchopneumonia.[12] The more frequent occurrence of adrenal hemorrhage with surgery on the biliary and gastrointestinal systems, especially when associated with intra-abdominal sepsis or cancer, is noteworthy.

The actual incidence of postoperative adrenal hemorrhage is unknown; many cases are never diagnosed and unilateral adrenal hemorrhage may not result in adrenal insufficiency. The incidence of adrenal hemorrhage in post-mortem examinations ranges from 0.14 to 0.69% (Table 2). Mitchell and Angrist[24] and Plaut[28] reported an incidence of adrenal hemorrhage of 0.6 and 9% respectively in consecutive post-mortem studies but the autopsy findings correlated poorly with the clinical findings in these patients.

Adrenal hemorrhage seems to primarily affect the aged, as all of our patients were 61 years of age or older. Greendyke[13] found that 70% of his patients were 56 or older and 36% were over 70 years of age. There was a marked predominence of men in our series, but previous reports suggest an equal incidence in men and women. In women, however, the mean age is often considerably younger because adrenal hemorrhage is frequently associated with a difficult pregnancy or delivery. Bilateral hemorrhage is most frequent in this group of patients. Botteri[5] found 14 of 23 cases to be bilateral, Berte,[3] 17 of 24 cases and Clark, et al.,[6] 4 of 7 cases. In contrast, Greendyke[13] found bilateral adrenal hemorrhage in only 33% of his patients. When unilateral adrenal hemorrhage occurs, it usually involves the right adrenal gland. Theories to explain why right-sided hemorrhage is more common are based upon its more vulnerable position in front of the spinal column and its direct venous drainage into the inferior vena cava. With bilateral adrenal hemorrhage, acute adrenal insufficiency rapidly develops. Symptoms of acute adrenal insufficiency are agitation, confusion, profound lethargy, emesis, fever and progressive asthenia with hypotension, cyanosis and tachycardia. In animals these changes occur in 2 to 3 days[17] following bilateral adrenalectomy. A similar temporal sequence occurs in patients with adrenal hemorrhage secondary to anticoagulation.[1] It also takes 2 to 3 days for the classical hyponatremia and hyperkalemia to develop. The symptoms resulting from adrenal hemorrhage may be due to acute adrenal insufficiency, retroperitoneal hemorrhage, or to both of these problems. In most cases the hemorrhage, contained within the retroperitoneum is

TABLE 2. *Incidence of Adrenal Hemorrhage (Unassociated with adrenal tumor of Waterhouse-Friderichsen syndrome)*

	Hemorrhage	No. of Autopsies	%
Botteri & Orell[5] (1964)	15	10852	.14
Greendyke[13] (1965)	33	21000	.16
Amador[1] (1965)	30	4325	.69
Jagatic[16] (1967)	12*	1793	.32
Fox[12] (1969)	5	1239	.40

*all bilateral

limited in amount, but massive hemorrhage has been described.[20] Of clinical importance is the fact that unilateral adrenal hemorrhage sometimes causes acute adrenal failure with an increase in the eosinophil count and a decrease in serum 17 hydroxy-and 17-ketosteroid concentration.[12,18] An eosinophil count of 50mm^3 or greater is considered to be highly suggestive of adrenal insufficiency in the acutely ill.[9]

Wilbur and Rich[36] were the first to demonstrate the experimental induction of adrenal tubular degeneration in rats by giving them pharmacological doses of ACTH. Rich[30] had previously observed focal adrenal necrosis in the adrenal glands of two patients with fulminating infection. Proctor and Rawson[29] found that corticosteroids increase the sensitivity of the adrenal cortex, leading to necrotic degeneration and massive bleeding. Shaberg[33] demonstrated small focal hemorrhages associated with necrotic cells in the adrenal cortex of guinea pigs treated with 30 mg of ACTH for 24 hours. Barta, Beregi and Jellinek[2] produced adrenal hemorrhage in rabbits by provoking insulin shock and simultaneously administering cortisone. Selye[31] found adrenal hemorrhage infrequently after operation in man but adrenal congestion and hemorrhage was more common in the cat. Levin and Cluff[21] produced adrenal hemorrhage by giving Thoratrast, ACTH, or endotoxin—all of which cause intensive adrenal stimulation. Mosley and Cluff[25] showed that these agents produced multiple coagulation defects and vascular-wall injury. Margaretten, et al.[23] produced adrenal hemorrhage in rats by disseminated intravascular clotting and inhibition of fibrinolysis. These studies suggest that both exogenous ACTH and endogenous ACTH due to severe stress may contribute to adrenal hemorrhage. Boguth, Langendorff and Tonutti[4] demonstrated the importance of the pituitary in adrenal hemorrhage since the histological changes induced in guinea pig adrenal glands by diphtheria toxin were prevented by previous hypophysectomy. Heparin and other anticoagulants predispose patients to adrenal hemorrhage. Heparin also inhibits aldosterone biosynthesis within the adrenal gland and may result in hyperkalemia without adrenal hemorrhage.[38]

Other factors contributing to adrenal hemorrhage are vitamin and nutritional deficiencies. Deficiency of pantothenic acid in rats and vitamin C in guinea pigs may cause adrenal hemorrhage and necrosis.[27,34] Operative procedures contribute to underlying nutritional deficiencies and also may cause iatrogenic injury to the adrenal glands.

The history of the patients presented here as well as previous reports and laboratory studies support the fact that postoperative adrenal hemorrhage is most likely to occur in the elderly, poorly nourished patient who has sepsis, cancer, and abdominal or intracranial operations or who is treated with anticoagulants or steroids. If the diagnosis of adrenal hemorrhage is considered in patients whose condition deteriorates unexpectedly postoperatively, more patients will be treated and survive. The diagnosis of adrenal hemorrhage after operation is a diagnosis of exclusion and must be considered in a patient whose condition deteriorates unexpectedly. This diagnosis is supported by leukocytosis, an elevated BUN, a low serum cortisol concentration, an elevated eosinophil count and hyperkalemia with hyponatremia, although treatment may be necessary before these laboratory changes develop. Treatment should include glucocorticoids and either increased saline or mineralocorticoids. Surgical exploration may or may not be necessary, depending on the severity of the hemorrhage.

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